UKUDLA Machine Learning Certificate

Introduction to Machine Learning for Agri-Food Systems

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UKUDLA
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3. Feature Engineering Framework

Learning Objectives

By the end of this session, you will be able to:

Data Preparation

- **Diagnose** missing values and determine when to impute vs. drop.
- **Detect and cap** extreme outliers using the Interquartile Range (IQR).
- **Perform** Exploratory Data Analysis (EDA) to find hidden relationships.

Feature Engineering

- **Execute** a 4-step framework for creating agronomically meaningful features.
- **Engineer** interaction features (e.g., Temperature-Humidity Index).
- **Engineer** ratio features (e.g., N:P Ratio).
- **Encode** categorical variables without falling into the Dummy Variable Trap.

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Session Roadmap

Part 1: Preparing the Data

1. The Smart Farming 2024 Dataset
2. Data Cleaning (Missing Values & Outliers)
3. Exploratory Data Analysis (EDA)

Part 2: Feature Engineering

4. The 4-Step Framework
5. Creating Interaction & Ratio Features
6. Categorical Encoding & Collinearity

Our Goal for Today

Transform a raw, messy agricultural dataset into a clean, mathematically enriched dataset ready for machine learning models.

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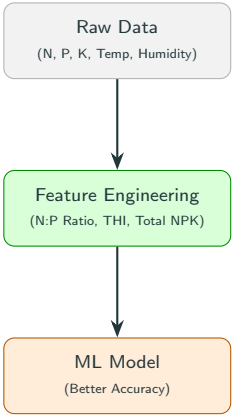
What is Feature Engineering?

The Definition

Feature engineering is the process of using **domain knowledge** to create new variables (features) from raw data to improve the performance of machine learning models.

The "Why"

Raw data often lacks the **interactions** and **relationships** that actually drive biological and agronomic outcomes.



"Machines find the math. You find the meaning."

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How to Engineer Features: A 4-Step Framework

Do not guess. Follow a structured process to create and validate new features.

Step 1: Domain Framing (The "Why")
Ask: What is the agronomic theory?
Example: High heat alone isn't bad, but high heat + high humidity causes crop failure.

Step 2: Transformation (The "How")
Ask: What is the mathematical formula?
Example: Create the Temperature-Humidity Index (THI).

Step 3: Selection (The "Filter")
Ask: Did this create redundancy?
Example: Check if the new THI feature is highly correlated ($|r| > 0.85$) with the raw Temperature feature.

Step 4: Validation (The "Test")
Ask: Did it actually work?
Example: Train the model. Did the new feature reduce the error (RMSE) on the test set?

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Feature Engineering Methods (Part 1)

1. Interaction Features

Combine two variables when their *joint* effect matters more than their individual effects.

Formula: $THI = (1.8T + 32) - (0.55 - 0.0055H)(1.8T - 26)$

Agronomic Rationale:
High temperature is stressful, but high humidity prevents the plant from cooling itself via transpiration. THI captures this combined heat stress.

2. Ratio Features

Capture the *balance* between two variables. Absolute amounts can be misleading.

Formula: $N_P_Ratio = \frac{N}{P+1}$

Agronomic Rationale:
A plant might have high absolute Nitrogen, but if Phosphorus is low, it cannot utilize the Nitrogen. The ratio matters more than the raw numbers.

Feature Engineering Methods (Part 2)

3. Aggregation Features

Combine multiple variables into a single summary metric.

Formula: $Total_NPK = N + P + K$

Agronomic Rationale:
Instead of looking at 3 separate nutrient values, this creates a "Total Fertility Index". It gives the model a quick snapshot of the overall nutrient availability.

4. Domain Indices

Use established scientific formulas to translate raw sensors into biological metrics.

Formula (GDD): $GDD = \sum \left(\frac{T_{max} + T_{min}}{2} - T_{base} \right)$

Agronomic Rationale:
Growing Degree Days measure the accumulated heat a crop receives. It predicts harvest timing much better than average temperature.

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The Dataset: Smart Farming 2024 (SF24)

2,201 farm records across 23 features.

- **Soil:** N, P, K, pH, Soil Type
- **Weather:** Temperature, Humidity, Rainfall
- **Management:** Irrigation, Fertilizer Usage
- **Targets:** Yield (Regression), Crop Label (Classification)

The Small Data Reality

With only 2,201 rows, we do not have "big data." If we feed 23 raw features into a model, it may overfit.

Feature engineering is how we extract maximum signal from limited agricultural data.

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Missing Values

The Problem: Sensors fail, farmers forget to log irrigation.

Why NOT the Mean?

If rainfall is $[0, 0, 0, 100]$, the mean is 25. But 25mm of rain never actually happened! The mean is highly sensitive to outliers.

Why the Median?

The median of $[0, 0, 0, 100]$ is 0. The median represents the "middle" reality of the field, making it robust and realistic.

Python Code: Median Imputation

```
df['Rainfall_mm'] = df['Rainfall_mm'].fillna(df['Rainfall_mm'].median())
```

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Outliers (IQR Method)

The Problem: Extreme weather events or logging errors can skew the model. We do not delete them—we **cap** them (Winsorization).

The Interquartile Range (IQR):

- $Q1 = 25\text{th percentile}$
- $Q3 = 75\text{th percentile}$
- $IQR = Q3 - Q1$
- **Upper Bound** $= Q3 + 1.5 \times IQR$
- **Lower Bound** $= Q1 - 1.5 \times IQR$

Python Code: Capping Outliers

```
Q1 = df['Rainfall'].quantile(0.25)
Q3 = df['Rainfall'].quantile(0.75)
IQR = Q3 - Q1
upper = Q3 + 1.5 * IQR
df['Rainfall'] = np.where(df['Rainfall'] >
upper,
upper, df['Rainfall'])
```

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Exploratory Data Analysis (EDA)

EDA is not about making pretty charts. It is about finding the missing link.

- **Correlation Matrices:** If N, P, and K are highly correlated with each other, the model will struggle. We call this **multicollinearity**.
- **Scatter Plots:** Plot 'Temperature' vs 'Yield'. Then plot 'Humidity' vs 'Yield'.
- **The Insight:** Temperature alone might have a weak correlation with Yield. Humidity alone might be weak. But together? *High heat + High humidity = Deadly fungal diseases and heat stress.*

The Cue for Engineering

When two variables interact to create an outcome that neither could produce alone, you must engineer a new feature to capture that interaction.

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The 4-Step Feature Engineering Process

1. Domain Framing (The "Why")
What agronomic theory explains this? *(e.g., Heat stress is exacerbated by humidity).*

2. Transformation (The "How")
Write the math. *(e.g., Create the THI index formula).*

3. Selection (The "Filter")
Check correlation. Does this new feature contain redundant info? *(Drop if* $|r| > 0.85$*).*

4. Validation (The "Test")
Train the model. Did the new feature reduce the Root Mean Squared Error (RMSE)?

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Engineering Interaction Features: THI

Domain Framing: A temperature of 35°C is dangerous to maize **only if** humidity is high (preventing transpiration).

Temperature-Humidity Index (THI) Formula

$$THI = (1.8 \times T + 32) - (0.55 - 0.0055 \times H) \times (1.8 \times T - 26)$$

Where T is Temperature (°C) and H is Humidity (%).

Agronomic Meaning:

- $THI < 72$: No heat stress.
- $THI 72 - 79$: Mild heat stress (yield reduction begins).
- $THI > 79$: Severe heat stress (crop failure risk).

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Engineering Ratio Features: Nutrient Balance

Domain Framing: Plants need Nitrogen (N) for leaves and Phosphorus (P) for roots. If you apply 100kg of N but only 1kg of P, the plant cannot utilize the N. The *ratio* matters more than the absolute amount.

The Formula

$$N_P_Ratio = \frac{N}{(P+1)}$$

We add 1 to the denominator to prevent division by zero if P is 0.

Other Agri-Food Ratios:

- **K:Na Ratio:** Determines salt tolerance in crops.
- **C:N Ratio:** Crucial for soil microbial activity and composting.

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Categorical Encoding

Models require numbers. They cannot calculate the distance between "Sandy" and "Clay".

One-Hot Encoding (Nominal)

Use when there is **no order**.

- Columns: `Soil_Sandy`, `Soil_Clay`
- A field is either 1 (True) or 0 (False).
- **Dummy Variable Trap:** Drop the first column. If it's not Sandy and not Clay, it *must* be Loam.

Ordinal Encoding (Ordered)

Use when there is a **hierarchy**.

- Growth Stage: Seedling (1), Vegetative (2), Flowering (3).
- Preserves the mathematical order of the categories.

The Danger of Too Many Features

Adding features increases complexity. If two features contain the same information, the model gets confused. This is **Multicollinearity**.

Correlation Matrix
If the absolute correlation $|r| > 0.85$, drop one of the two features.

VIF (Variance Inflation Factor)
A statistical measure. $VIF > 5$ indicates high collinearity.

Data: `session03_feature_engineering.ipynb`

What we will do together:

- Impute missing rainfall data.
- Calculate the Temperature-Humidity Index (THI).
- Calculate the N:P Ratio.
- One-hot encode Soil Type.
- Drop highly correlated features.

Key Takeaways

1. Domain knowledge drives feature engineering. Machines find math; you find meaning.

2. Always justify a new feature with agronomic theory (e.g., heat stress, nutrient balance).

3. More features $\neq$ better models. Check for collinearity ($|r| > 0.85$).

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